UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

a) List TWO (2) advantages of SMP organization.
b) Define cluster organization.
c) Give ONE (1) benefit of cluster organization.

Source: CSC429 (7).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Convert 2.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: 2.510 = 0x40200000 (sign=0, exponent=10000000, fraction=01000000000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A3 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Computer generations and history

Identify the correct generation based on the following descriptions.
a)
The production of personal computer marks the main feature of this generation. The price of computer during this period is the lowest compared to the other generation. It provides fast processing ability with less power consumption.
b)
Machines that emulate human intelligence which can process and respond to natural language is developed in this generation.
c)
In this generation, air condition is no longer required because the heat produced by the machine is reduced. However, fan is needed for heat discharging.
d)

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Integrated circuit comprises of a number of transistors, resistors and capacitors are invented by Jack Kilby in this generation.
e)
Assembly language is introduced during this generation to replace machine language for more reliable and improved performance.

Source: CSC429 (4).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

a) Why input/output problem occurs?
b) Describe THREE (3) categories for input/output module and provide an appropriate example each.

Source: CSC429 (6).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Input/output and interrupts

a) When is an I/O module installed?
b) What will happen if the I/O module is not installed?
c) Explain ONE (1) category of I/O module
d) What does it mean by 'hot connection'?
e) List THREE (3) common interfaces to external devices.

Source: CSC429 (21).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: -35 - 24

a) Calculate -35 - 24 using sign and magnitude.

Answer: left -35 -> 1 0100011; right 24 -> 0 0011000; -35 - 24 = -59 -> sign-magnitude 1 0111011

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate -35 - 24 using 1's and 2's complement.

Answer: left -35 -> 1's 11011100, 2's 11011101; right 24 -> 1's 00011000, 2's 00011000; -35 - 24 = -59 -> 1's 11000100, 2's 11000101

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

Computer consists of four main components as illustrated in Figure 1. Each component has its own function in ensuring that the computer system works well. Question 4 is based on Figure 1.
Peripherals
Central

Main
Processing
Memory
Unit
Computer
Systems
Interconnection
Input
Output
Communication
lines

Figure 1 : Components of computer

a) Explain THREE (3) types of system interconnection.
b) State TWO (2) sub-components of central processing unit and its function.

Source: CSC429 (3).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Memory hierarchy and cache

a) What is a cache?
b) Elaborate on what will happen if there is no cache in the computer system,
c) Comparing L1 and L2 cache, discuss which cache would have faster speed with relevant justification.
d) Discuss TWO (2) policies to write data to the cache to ensure consistency between the cache and the main memory (RAM).

Source: CSC429 (18).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Computer architecture and CPU organisation

a) Draw a complete machine cycle by giving the exact location of each phase.
b) Briefly, describe each phase in the machine cycle.

Source: CSC429 (8).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.
y = A(A + 2B/C-D 2)
Figure 2
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.
Table 1: Machine Instruction
One-address
Three-address
Zero-address
PUSHp
LOADp
ADD (p,q,r)
POPp
STORp
SUB (p,q,r)
ADD
ADDp
MUL (p.q.r)
SUB
SUBp
DIV(p,q,r)
MUL
MULp
DIV
DIVp

Source: CSC429 (17).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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